

Research Project
Injectivity and quasi-injectivity of products of some polynomials
and Dedekind psi function

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Chapter 1

Introduction

Dedekind psi function ψ is an arithmetic function defined by, $\psi(n) = n \prod_{p|n} (1 + \frac{1}{p})$ for all $n \in \mathbb{N}$, where the product is taken over all primes p dividing n and $\psi(1) = 1$. This function is injective because $\psi(4) = 6 = \psi(5)$ and $\psi(6) = 12 = \psi(8)$. It has a similar expression to Euler's totient function ϕ with $\phi(n) = n \prod_{p|n} (1 - \frac{1}{p})$ for all $n \in \mathbb{N}$ where the product is taken over all primes p dividing n . Obviously, ϕ is not injective because $\phi(3) = \phi(4) = \phi(6) = 2$ and $\phi(5) = \phi(8) = \phi(10) = 4$. In 2011, James noticed that the function $F : \mathbb{N} \rightarrow \mathbb{N}$ defined by $F(n) = n\phi(n)$ may be injective. He proposed a question asking whether F is injective on the website Mathematics Stack Exchange [1]. There are many mathematicians answered the this question and also gave the proof. This motivate us to study the injectivity of product of some polynomial functions and Dedekind psi function.

Later in 2021, P. prapanpong [2] defined the definition of quasi-injective function. He studied quasi-injectivity of some arithmetic functions. such as tau function, Jordan totient function, sigma function, Möbius function, prime divisor function, distinct prime divisor function and Alladi-Erdős function. He showed that tau function, Jordan totient function and sigma function are quasi-injective. On the other hand, prime divisor function, distinct prime divisor function and Alladi-Erdős function are not quasi-injective. One can easily see all injective functions are also quasi-injective. So we would like to further investigate some products of some polynomials and Dedekind psi function that may be quasi-injective although they are not injective.

In this research, we study about injectivity and quasi-injectivity of product of some polynomials and Dedekind psi function. We found that, for each $m \in \mathbb{N}$ the function $F : \mathbb{N} \rightarrow \mathbb{N}$ defined by $F(n) = n^m \psi(n)$ for all $n \in \mathbb{N}$ is injective where $m \in \mathbb{N}$. We also found that ψ and the function $F : \mathbb{N} \rightarrow \mathbb{N}$ defined by $F(n) = (n + c)\psi(n)$, where $c > 0$ for all $n \in \mathbb{N}$, are not injective but quasi-injective.

Chapter 2

Preliminaries

Definition 2.1. A function $f : A \rightarrow B$ is **injective**, if $f(x_1) = f(x_2)$ implies $x_1 = x_2$ for all $x_1, x_2 \in A$.

Definition 2.2. [2] A function $f : \mathbb{N} \rightarrow \mathbb{N}$ is **quasi-injective** if, for all $a, b \in \mathbb{N}$, the condition $f(an) = f(bn)$ for all $n \in \mathbb{N}$ implies $a = b$. In addition, if $f : \mathbb{N} \rightarrow \mathbb{N}$ and $\ell \in \mathbb{N}$ then f is **quasi-injective of order ℓ** if $f^{(1)} f^{(2)} f^{(3)} \dots f^{(\ell)}$ are quasi-injective, i.e., for $a, b, k \in \mathbb{N}$ with $1 \leq k \leq \ell$ if $f^{(k)}(an) = f^{(k)}(bn)$ for all $n \in \mathbb{N}$, then $a = b$.

Example 2.3. [2], There is a function which is not quasi-injective. For each $n \in \mathbb{N}$, let $\Omega(n)$ be the number of prime divisors of n counted with multiplicities. The function Ω is completely additive, that is $\Omega(mn) = \Omega(m) + \Omega(n)$ for all $m, n \in \mathbb{N}$. Let $k \geq 2$, $p_1, p_2, \dots, p_k$ be distinct primes, $c = p_1 p_2 \dots p_k$, $a = p_1 c$ and $b = p_2 c$. Then $\Omega(a) = \Omega(b)$. As Ω is completely additive, $\Omega(an) = \Omega(a) + \Omega(n) = \Omega(b) + \Omega(n) = \Omega(bn)$ for all $n \in \mathbb{N}$. However, $a \neq b$. Thus Ω is not quasi-injective.

Theorem 2.4. If $f : \mathbb{N} \rightarrow \mathbb{N}$ is injective, then f is quasi-injective.

Proof. Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be an injective function. Let $a, b \in \mathbb{N}$. Assume that $f(an) = f(bn)$ for all $n \in \mathbb{N}$. Then $f(a) = f(a \cdot 1) = f(b \cdot 1) = f(b)$. Since f is injective, we have that $a = b$. Therefore, f is quasi-injective. $\square$

Remark As a result of Theorem 2.4, if $f : \mathbb{N} \rightarrow \mathbb{N}$ is injective then f is quasi-injective of all order in $\mathbb{N}$.

In general, a quasi-injective function may be not injective. We will show this later.

Theorem 2.5. (*The Fundamental Theorem of Arithmetic*) A positive integer $n > 1$ can be expressed uniquely as a product of primes,

$$n = p_1^{a_1} p_2^{a_2} p_3^{a_3} \dots p_k^{a_k}$$

where $p_1, p_2, p_3, \dots, p_k$ are distinct prime numbers and a_i are positive integers for all $1 \leq i \leq k$.

In number theory, the Dedekind psi function $\psi : \mathbb{N} \rightarrow \mathbb{N}$ is defined by

$$\psi(n) = n \prod_{p|n} \left(1 + \frac{1}{p}\right)$$

where the product is taken over all primes p dividing n and $\psi(1) = 1$.

Example 2.6. We can calculate the value of $\psi(n)$ where $n \in \mathbb{N}$ as shown below.

$$\psi(4) = 4\left(1 + \frac{1}{2}\right) = 6 = 5\left(1 + \frac{1}{5}\right) = \psi(5),$$

$$\psi(6) = 6\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{3}\right) = 12 = 9\left(1 + \frac{1}{3}\right) = \psi(9).$$

According to [3], ψ is multiplicative, that is $\psi(1) = 1$ and $\psi(ab) = \psi(a)\psi(b)$ whenever a and b are coprimes.

Example 2.7. A simple calculation shows that

$$\psi(15) = 24 = \psi(3)\psi(5),$$

$$\psi(12) = 24 = \psi(3)\psi(4),$$

$$\text{and } \psi(12) = 24 \neq 36 = \psi(2)\psi(6),$$

because $\gcd(2, 6) = 2 \neq 1$.

Chapter 3

Injectivity and quasi-injectivity of Dedekind psi function

This section we will show injectivity and quasi-injectivity of Dedekind psi function. It is well-known that ψ is not injective as show in the table 4.1. However, it is quasi-injective shown as follows.

Theorem 3.1. *Let $a, b, c, k, m \in \mathbb{N}$. Then the following statements hold*

- (i) *If $a|b$ then $\psi(ab) = a\psi(b)$.*
- (ii) *If $a|c$ and $b|c$ then $b\psi(ac) = a\psi(bc)$.*
- (iii) *If $a|b$ then $\psi^{(m)}(a)|\psi^{(m)}(b)$.*
- (iv) *$\psi(a^k b) = a^{(k-1)}\psi(ab)$.*
- (v) *If $k \geq m + 1$, then $\psi^{(m)}(a^k b) = a\psi^{(m)}(a^{k-1}b)$.*
- (vi) *If $k \geq m + 1$, then $a|\psi^{(m)}(a^k)$.*
- (vii) *If $k \geq m + 1$ and $c \geq m + 1$, then $a|\psi^{(m)}(a^k b^c)$ and $b|\psi^{(m)}(a^k b^c)$.*

Proof. (i) Let $a|b$. For any prime number p , we have that $p|ab$ if and only if $p|b$. So

$$\begin{aligned}\psi(ab) &= (ab) \prod_{p|ab} \left(1 + \frac{1}{p}\right) \\ &= a \left(b \prod_{p|b} \left(1 + \frac{1}{p}\right)\right) \\ &= a\psi(b).\end{aligned}$$

(ii) Assume that $a|c$ and $b|c$. From (i), we have that

$$\begin{aligned} b\psi(ac) &= ab\psi(c) \\ &= a\psi(bc). \end{aligned}$$

(iii) Assume that $a|b$. If $a = 1$, then $\psi^{(m)}(1) = 1$. Assume $a \geq 2$. Let $a = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$ and $b = b_0 p_1^{b_1} p_2^{b_2} \dots p_k^{b_k}$ where $p_1 < p_2 < \dots < p_k$ are prime numbers, $a_i, b_i, b_0 \in \mathbb{N}$, and $b_i \geq a_i$ for all $i = 1, 2, \dots, k$, and $\gcd(b_0, p_1, p_2, \dots, p_k) = 1$.

$$\begin{aligned} \psi(a) &= \prod_{i=1}^k (p_i^{(a_i-1)}) \prod_{i=1}^k (p_i + 1). \\ \psi(b) &= \prod_{i=1}^k (p_i^{(b_i-1)}) \prod_{i=1}^k (p_i + 1) \psi(b_0). \end{aligned}$$

Therefore, $\psi(a)|\psi(b)$. Then

$$a|b \Rightarrow \psi(a)|\psi(b) \Rightarrow \psi^{(2)}(a)|\psi^{(2)}(b) \Rightarrow \dots \Rightarrow \psi^{(m)}(a)|\psi^{(m)}(b).$$

Therefore, (iii) is true for all $m \geq 1$.

(iv) This is a consequence of (i).

(v) We will proof by mathematical induction on m . Let $m = 1$ and $k \geq 2$. By (i),

$$\begin{aligned} \psi(a^k b) &= \psi(a(a^{k-1}b)) \\ &= a\psi(a^{k-1}b). \end{aligned}$$

Next, let $m \geq 2$ and assume that if $k \geq m+1$, then $\psi^{(m)}(a^k b) = a\psi^{(m)}(a^{k-1}b)$. Let $k \geq m+2$.

$$\begin{aligned} \psi^{(m+1)}(a^k b) &= \psi(\psi^{(m)}(a^k b)) \\ &= \psi(a\psi^{(m)}(a^{k-1}b)). \end{aligned} \tag{3.1}$$

Since $k-1 \geq m+1$, we have $\psi^{(m)}(a^{k-1}b) = a\psi^{(m)}(a^{k-2}b)$. This implies that $a|\psi^{(m)}(a^{k-1}b)$. By (i), we have $\psi^{(m+1)}(a^k b) = \psi(a\psi^{(m)}(a^{k-1}b)) = a(\psi(\psi^{(m)}(a^{k-1}b))) = a(\psi^{(m+1)}(a^{k-1}b))$.

(vi) Let $k \geq m+1$. From (v), we have $\psi^{(m)}(a^k) = a\psi^{(m)}(a^{k-1})$. Therefore, $a|\psi^{(m)}(a^k)$.

(vii) Let $k \geq m+1$ and $c \geq m+1$. From (iii) and (vi), we have that

$$\begin{aligned} a|\psi^{(m)}(a^k) \quad \text{and} \quad \psi^{(m)}(a^k)|\psi^{(m)}(a^k b^c), \\ b|\psi^{(m)}(b^c) \quad \text{and} \quad \psi^{(m)}(b^c)|\psi^{(m)}(a^k b^c). \end{aligned}$$

Thus $a|\psi^{(m)}(a^k b^c)$ and $b|\psi^{(m)}(a^k b^c)$. □

Theorem 3.2. Let $a, b, m \in \mathbb{N}$ and $\psi^{(m)}(an) = \psi^{(m)}(bn)$ for all $n \in \mathbb{N}$. Then $a = b$. In other words, ψ is quasi-injective of any order.

Proof. Since $a^m b^m \in \mathbb{N}$, we have that $\psi^{(m)}(a^{m+1} b^m) = \psi^{(m)}(b^{m+1} a^m)$. From Theorem 3.1 (v), we have

$$\psi^{(m)}(a^{m+1} b^m) = \psi^{(m)}(a(a^m b^m)) = a\psi^{(m)}(a^m b^m),$$

and

$$\psi^{(m)}(b^{m+1} a^m) = \psi^{(m)}(b(b^m a^m)) = b\psi^{(m)}(a^m b^m).$$

Therefore, $a\psi^{(m)}(a^m b^m) = b\psi^{(m)}(a^m b^m)$, and so $a = b$. □

As a result of Theorem 3.2, a quasi-injective function is not necessary injective.

Chapter 4

Injectivity and quasi-injectivity of products of some polynomials and Dedekind psi function

Lemma 4.1. *Let $m, r \in \mathbb{N}$ and $\alpha, \beta \in \mathbb{N} \cup \{0\}$. If $m^r \psi(m) = (2^\alpha 3^\beta)^r \psi(2^\alpha 3^\beta)$, then $m = 2^\alpha 3^\beta$.*

Proof. Assume that $m^r \psi(m) = (2^\alpha 3^\beta)^r \psi(2^\alpha 3^\beta)$.

Case1 $\alpha = 0$ and $\beta = 0$. Then $m^r \psi(m) = 1^r \psi(1) = 1$, and so $m = 1$.

Case2 $\alpha > 0$ and $\beta = 0$. Then $m^r \psi(m) = (2^\alpha)^r \psi(2^\alpha) = 2^{\alpha r + \alpha - 1} \cdot 3$. Thus $m = 2^k 3^t$ where $k, t \in \mathbb{N} \cup \{0\}$. Suppose, for a contradiction, that $t \neq 0$.

Case2.1 $m = 2^k 3^t$. Then $2^{\alpha r + \alpha - 1} \cdot 3 = m^r \psi(m) = 2^{kr + k - 1} \cdot 3^{tr + t}$. By uniqueness of prime factorization, $1 = tr + t > 1$, a contradiction.

Case2.2 $m = 3^t$. Then $2^{\alpha r + \alpha - 1} \cdot 3 = m^r \psi(m) = 3^{tr + t - 1} \cdot 2^2$. By uniqueness of prime factorization, we have $tr + t - 1 = 1$ and $\alpha r + \alpha - 1 = 2$. As $tr + t = 2$, we have $t = 1$ and $r = 1$. But $\alpha = \frac{3}{2}$, a contradiction.

Therefore $t = 0$, and so $m = 2^k$. Thus $2^{\alpha r + \alpha - 1} \cdot 3 = m^r \psi(m) = 2^{kr + k - 1} \cdot 3$. By uniqueness of prime factorization, $\alpha r + \alpha - 1 = kr + k - 1$, i.e., $\alpha = k$. This implies that $m = 2^\alpha = 2^\alpha 3^\beta$.

Case3 $\alpha = 0$ and $\beta > 0$. Then $m^r \psi(m) = (3^\beta)^r \psi(3^\beta) = 3^{\beta r + \beta - 1} \cdot 2^2$. Thus $m = 2^k 3^t$ where $k, t \in \mathbb{N} \cup \{0\}$. Suppose, for a contradiction, that $k \neq 0$.

Case3.1 $m = 2^k 3^t$. Then $3^{\beta r + \beta - 1} \cdot 2^2 = m^r \psi(m) = 3^{tr + t} \cdot 2^{kr + k + 1}$. By uniqueness of prime factorization, $kr + k + 1 = 2$, and so $2 \leq kr + k = 1$, a contradiction.

Case3.2 $m = 2^k$. Then $3^{\beta r + \beta - 1} \cdot 2^2 = m^r \psi(m) = 2^{kr + k - 1} \cdot 3$. By uniqueness of prime factorization, $\beta r + \beta - 1 = 1$, and $kr + k - 1 = 2$. Since $\beta r + \beta = 2$, we have $\beta = 1$ and

$r = 1$. As $kr + k - 1 = 2$, we have that $k = \frac{3}{2}$, a contradiction.

Therefore $k = 0$, and so $m = 3^t$. Thus $3^{\beta r + \beta - 1} \cdot 2^2 = m^r \psi(m) = 3^{tr + t - 1} \cdot 2^2$. By uniqueness of prime factorization, $\beta r + \beta - 1 = tr + t - 1$, i.e., $\beta = t$. This implies that $m = 3^\beta = 2^\alpha 3^\beta$.

Case4 $\alpha > 0$ and $\beta > 0$. Then $m^r \psi(m) = (2^\alpha 3^\beta)^r \psi(2^\alpha 3^\beta) = 2^{\alpha r + \alpha + 1} \cdot 3^{\beta r + \beta}$. This implies that $m = 2^k 3^t$ where $k, t \in \mathbb{N} \cup \{0\}$. Suppose, for a contradiction, that $k = 0$ or $t = 0$.

Case4.1 $k = 0$. Then $2^{\alpha r + \alpha + 1} \cdot 3^{\beta r + \beta} = m^r \psi(m) = 3^{tr + t - 1} \cdot 2^2$. By uniqueness of prime factorization, $3 \leq \alpha r + \alpha + 1 = 2$, a contradiction.

Case4.2 $t = 0$. Then $2^{\alpha r + \alpha + 1} \cdot 3^{\beta r + \beta} = m^r \psi(m) = 2^{kr + k - 1} \cdot 3$. By uniqueness of prime factorization, $1 = \beta r + \beta \geq 2$, a contradiction.

So $k \neq 0$, $t \neq 0$, and $m = 2^k 3^t$. So $2^{\alpha r + \alpha + 1} \cdot 3^{\beta r + \beta} = m^r \psi(m) = 2^{kr + k + 1} 3^{tr + t}$.

By uniqueness of prime factorization, we have $\alpha r + \alpha + 1 = kr + k + 1$ and $\beta r + \beta = tr + t$, i.e., $\alpha = k$ and $\beta = t$. This implies that $m = 2^\alpha 3^\beta$. $\square$

Theorem 4.2. Let $m \in \mathbb{N}$. The function $F : \mathbb{N} \rightarrow \mathbb{N}$ defined by $F(n) = n^m \psi(n)$ is injective.

Proof. Let $x, y, m \in \mathbb{N}$. Suppose that $F(x) = F(y)$.

Case1 $x = 1$ or $y = 1$. Without loss of generality, let $x = 1$. This implies that $1 = x^m \psi(x) = y^m \psi(y)$. So $y = 1$. Otherwise, $y^m \psi(y) \geq 2 \cdot 1 = 2$. Therefore, $x = y$.

Case2 $x \neq 1$ and $y \neq 1$. By prime factorization of x and y , let

$$x = p_1^{\alpha_1} p_2^{\alpha_2} p_3^{\alpha_3} \dots p_k^{\alpha_k}$$

$$y = q_1^{\beta_1} q_2^{\beta_2} q_3^{\beta_3} \dots q_\ell^{\beta_\ell}$$

where $p_1 < p_2 < p_3 < \dots < p_k$ and $q_1 < q_2 < q_3 < \dots < q_\ell$, $\alpha_i > 0$ and $\beta_j > 0$ for all $1 \leq i \leq k$ and $1 \leq j \leq \ell$. Since $F(x) = F(y)$ and ψ is multiplicative, we have

$$\prod_{i=1}^k [p_i^{\alpha_i(m+1)-1} (p_i + 1)] = \prod_{j=1}^{\ell} [q_j^{\beta_j(m+1)-1} (q_j + 1)]. \quad (4.1)$$

Case 2.1 $p_k \leq 3$ or $q_\ell \leq 3$. Then Lemma 4.1 implies that $x = y$

Case 2.2 $p_k \geq 5$ and $q_\ell \geq 5$. Then the largest prime factor of LHS of (4.1) is p_k because $\frac{p_i + 1}{2} < p_k$ for all $1 \leq i \leq k$. Similarly, q_ℓ is the largest prime factor of RHS of (4.1). Then by uniqueness of prime factorization, that $p_k = q_\ell$ and $\alpha_k = \beta_\ell$. Hence (4.1) is reduced to

$$\prod_{i=1}^{k-1} [p_i^{\alpha_i(m+1)-1} (p_i + 1)] = \prod_{j=1}^{\ell-1} [q_j^{\beta_j(m+1)-1} (q_j + 1)].$$

By continually processing the similar argument as above, we can conclude that $k = \ell, p_i = q_j$ and $\alpha_i = \beta_j$ for all $1 \leq i \leq k$. Hence $x = y$. Therefore, F is injective. $\square$

Remark As a result of Theorem 2.2, F defined in Theorem 4.2 is also quasi-injective.

We further investigate the products of the polynomial function $n \rightarrow n + c$, where c is an integer, and the Dedekind psi function. We found that their product is not injective as shown in Table 4.2 to 4.7. However it is quasi-injective when $c > 0$ as shown follows.

Theorem 4.3. *Let $c > 0$. The function $F : \mathbb{N} \rightarrow \mathbb{N}$ defined by $F(n) = (n + c)\psi(n)$ is quasi-injective.*

Proof. Let $a, b \in \mathbb{N}$ and $F(an) = F(bn)$ for all $n \in \mathbb{N}$. Thus $a = b$. By choosing $n = ab$, we have that

$$\begin{aligned} (a^2b + c)\psi(a^2b) &= (ab^2 + c)\psi(ab^2) \\ \text{Thus } (a^3b + ac)\psi(ab) &= (ab^3 + bc)\psi(ab) \\ a^3b + ac &= ab^3 + bc \\ ac - bc &= ab^3 - a^3b \\ (a - b)c &= ab(b^2 - a^2) \\ (a - b)c &= ab(b - a)(a + b). \end{aligned}$$

Suppose, for a contradiction, that $a \neq b$. Without loss of generality, let $a > b$. Then $(a - b)c > 0$, whereas $ab(b - a)(a + b) < 0$, contradiction. Therefore $a = b$ and F is quasi-injective. $\square$

Table 4.1: The table shows the values $n_1, n_2 \in [1, 100]$ such that $\psi(n_1) = \psi(n_2)$ where ψ is the Dedekind psi function

n_1	n_2	$\psi(n_1) = \psi(n_2)$
4	5	6
6	8	12
6	9	12
6	11	12
8	9	12
8	11	12
9	11	12
10	17	18
12	14	24
12	15	24
12	16	24
12	23	24
14	15	24
14	16	24
14	23	24
15	16	24
15	23	24
16	23	24
18	20	36
18	22	36
18	27	36
20	22	36
20	27	36
21	31	32
22	27	36
24	28	48
24	32	48

n_1	n_2	$\psi(n_1) = \psi(n_2)$
24	33	48
24	35	48
24	47	48
25	29	30
26	41	42
28	32	48
28	33	48
28	35	48
28	47	48
30	36	72
30	40	72
30	44	72
30	45	72
30	46	72
30	51	72
30	55	72
30	71	72
32	33	48
32	35	48
32	47	48
33	35	48
33	47	48
34	53	54
35	47	48
36	40	72
36	44	72
36	45	72

n_1	n_2	$\psi(n_1) = \psi(n_2)$
36	46	72
36	51	72
36	55	72
36	71	72
38	59	60
39	49	56
40	44	72
40	45	72
40	46	72
40	51	72
40	55	72
40	71	72
42	48	96
42	56	96
42	62	96
42	63	96
42	64	96
42	69	96
42	77	96
44	45	72
44	46	72
44	51	72
44	55	72
44	71	72
45	46	72
45	51	72
45	55	72

n_1	n_2	$\psi(n_1) = \psi(n_2)$
45	71	72
46	51	72
46	55	72
46	71	72
48	56	96
48	62	96
48	63	96
48	64	96
48	69	96
48	77	96
50	58	90
50	89	90
51	55	72
51	71	72
52	65	84
52	83	84
54	68	108
54	81	108
54	85	108
55	71	72
56	62	96
56	63	96
56	64	96
56	69	96
56	77	96
57	79	80
58	89	90

n_1	n_2	$\psi(n_1) = \psi(n_2)$
60	66	144
60	70	144
60	72	144
60	80	144
60	88	144
60	92	144
60	94	144
60	99	144
62	63	96
62	64	96
62	69	96
62	77	96
63	64	96
63	69	96
63	77	96
64	69	96
64	77	96
65	83	84
66	70	144
66	72	144
66	80	144
66	88	144
66	92	144
66	94	144
66	99	144
68	81	108
68	85	108
69	77	96
70	72	144

n_1	n_2	$\psi(n_1) = \psi(n_2)$
70	80	144
70	88	144
70	92	144
70	94	144
70	99	144
72	80	144
72	88	144
72	92	144
72	94	144
72	99	144
75	76	120
75	87	120
75	95	120
76	87	120
76	95	120
78	98	168
80	88	144
80	92	144
80	94	144
80	99	144
81	85	108
84	96	192
87	95	120
88	92	144
88	94	144
88	99	144
92	94	144
92	99	144
94	99	144

Table 4.2: Let $F : \mathbb{N} \rightarrow \mathbb{N}$ defined by $F(n) = (n+1)\psi(n)$. This table shows the values $n_1, n_2 \in [1, 10000]$ such that $F(n_1) = F(n_2)$

n_1	n_2	$F(n_1) = F(n_2)$
80	107	11664
140	187	40608
322	407	186048
494	524	415800
510	656	662256
608	695	584640
699	779	655200
948	1167	1822080
960	1487	2214144
1178	1309	2263680
1215	1295	2363904
1232	1643	2840832
1518	2231	5249664
1615	1817	3490560
2304	2765	10621440
2344	2813	8273160
2444	2933	9858240
2610	3356	16919280
2794	3353	12879360

n_1	n_2	$F(n_1) = F(n_2)$
2924	3509	13899600
2925	3134	14747040
3069	3683	14146560
3486	5071	28119168
3835	4109	19333440
4080	5935	42311808
4479	4979	26772480
4558	4652	32496552
4640	6187	40098240
5345	5777	34321320
6408	8873	83060640
6696	8687	92579328
6768	7735	93574656
6915	7006	76684608
7264	8717	79508160
7292	7955	79814592
7505	8339	72057600
8378	8504	108591840
8590	9158	132988680

Table 4.3: Let $F : \mathbb{N} \rightarrow \mathbb{N}$ defined by $F(n) = (n + 2)\psi(n)$. This table shows the values $n_1, n_2 \in [1, 10000]$ such that $F(n_1) = F(n_2)$

n_1	n_2	$F(n_1) = F(n_2)$
84	127	16512
110	124	24192
118	133	21600
150	188	54720
448	478	345600
606	644	744192
754	943	1179360
880	1006	1524096
988	1048	1663200
1020	1312	2649024
1274	1419	3001152
1458	1969	4257360
1494	2077	4523904
2430	2590	14183424
2734	3247	11228544
3304	3478	19042560
3328	3358	17902080
3386	3991	17217816
3724	4345	25038720
3798	5033	29001600
3880	4527	27391392
4968	5678	51528960
6048	7258	83635200
7456	9151	83768256
7670	8218	116000640
9116	9304	129986208

Table 4.4: Let $F : \mathbb{N} \rightarrow \mathbb{N}$ defined by $F(n) = (n + 3)\psi(n)$. This table shows the values $n_1, n_2 \in [1, 10000]$ such that $F(n_1) = F(n_2)$

n_1	n_2	$F(n_1) = F(n_2)$
12	17	360
60	81	9072
72	87	10800
198	265	86832
240	321	139968
369	431	187488
372	497	288000
420	561	487296
508	581	392448
600	801	868320
624	833	842688
927	1037	1160640
966	1221	2232576
1376	1573	2912448
1482	1572	4989600

n_1	n_2	$F(n_1) = F(n_2)$
1725	1917	4976640
2272	2337	7862400
2272	2597	7862400
2337	2597	7862400
2490	3321	15077664
2844	3501	16398720
3534	3927	27164160
3696	4929	34089984
3854	4405	23327136
4422	5012	43329600
4572	5487	42163200
4845	5451	41886720
5838	6192	78503040
6844	8347	73947600
6888	9185	111138048

Table 4.5: Let $F : \mathbb{N} \rightarrow \mathbb{N}$ defined by $F(n) = (n-1)\psi(n)$. This table shows the values $n_1, n_2 \in [1, 10000]$ such that $F(n_1) = F(n_2)$

n_1	n_2	$F(n_1) = F(n_2)$
16	19	360
40	53	2808
286	379	143640
408	593	351648
477	505	308448
708	1009	1018080
783	919	844560
846	1015	1460160
946	1189	1496880
1001	1121	1344000
1023	1169	1569792
1136	1363	1961280
1344	1897	4125696
1485	1909	3846528
1496	1795	3875040
1704	2097	5885568
1900	2533	6836400
2465	2733	7983360
2500	2626	11245500
2996	3595	15526080
3136	4033	16843760

n_1	n_2	$F(n_1) = F(n_2)$
3166	3799	15040080
3484	4387	19894896
3745	4213	19408896
3905	4393	20238336
4775	4961	27498240
5124	6759	60984192
5365	5663	36689760
5385	6731	46517760
5625	7031	50616000
6225	7781	62737920
6346	7897	63957600
6350	7257	73140480
6432	7413	83963136
6462	7456	83734560
6462	8947	83734560
7456	8947	83734560
6784	7345	70326144
7316	7701	84268800
7381	7627	60397920
7533	8065	78091776

Table 4.6: Let $F : \mathbb{N} \rightarrow \mathbb{N}$ defined by $F(n) = (n - 2)\psi(n)$. This table shows the values $n_1, n_2 \in [1, 10000]$ such that $F(n_1) = F(n_2)$

n_1	n_2	$F(n_1) = F(n_2)$
38	47	2160
72	82	10080
106	119	16848
132	158	37440
200	218	71280
326	371	159408
442	464	332640
452	477	307800
534	572	574560
552	602	633600
600	899	861120
780	1169	1568448
850	956	1373760
900	1349	1939680
954	1010	1850688
1294	1541	2511648
1352	1402	2948400
1440	2159	4969728
1503	1661	3026016
1566	1838	5067360
1716	2573	6910848
1802	2189	5248800
1804	1906	5449248
2002	2242	8064000
2046	2338	9418752

n_1	n_2	$F(n_1) = F(n_2)$
2888	2962	13160160
2970	3818	23079168
3406	3544	18871776
3940	4835	28070064
3948	5921	36366336
4150	5357	31358880
4232	4322	28019520
4488	6731	46510248
4838	6047	36560160
4930	5546	47900160
5000	5252	44982000
5148	5978	62246016
5174	6467	43444800
5874	6608	76101120
6114	6496	74810880
6195	6758	71343360
6195	8447	71343360
6758	8447	71343360
6240	9359	100606464
7152	8127	102960000
7490	8426	116453376
7790	9737	117754560
7810	8736	187815936
7850	8613	111598560
9550	9922	164989440

Table 4.7: Let $F : \mathbb{N} \rightarrow \mathbb{N}$ defined by $F(n) = (n - 3)\psi(n)$. This table shows the values $n_1, n_2 \in [1, 10000]$ such that $F(n_1) = F(n_2)$

n_1	n_2	$F(n_1) = F(n_2)$
42	55	3744
48	57	4320
120	159	33696
312	415	207648
480	639	549504
649	683	465120
819	955	1096704
843	943	947520
858	1137	1723680
861	927	1153152
1203	1343	1929600
1713	1979	3912480
2163	2563	7188480
2448	2937	12674880
2576	2979	11856384
2640	3519	18226944
2698	2943	11642400
2838	3567	17962560
2856	3807	19719936
2874	3483	16536960
2880	4387	19885824
3003	3363	16128000
3160	3693	18184320
3234	5029	26054784
3408	4089	23535360
3488	4103	18400800

n_1	n_2	$F(n_1) = F(n_2)$
3762	5015	32477760
4248	5663	36676800
4275	4803	30758400
4488	5385	46500480
4662	6215	50988096
4800	6399	55261440
4876	5319	37892448
5112	6291	52970112
5292	7055	63975744
5415	6235	49357440
5700	7599	82036800
5715	7171	52641792
5970	7803	85924800
6423	7279	55006560
6747	7871	65713536
6747	8447	65713536
7871	8447	65713536
6916	7808	92910720
6975	8963	80317440
7395	8319	95800320
7500	7878	134946000
7750	9147	111556800
7758	9309	120605760
7990	8805	124213824
8607	9085	104624640

Appendix

This command is used to calculate for the Dedekind psi function.

```
function m=D(n)
    if n==1
        m=1;
    else
        x=unique(factor(n));
        m=(n)*prod(1+1./(x)) ;
    end
```

This command is used to find $F(n) = (n+1)\psi(n)$ which is injective.

```
x=[];
for i=1:100000
    x=[x,sym((i+1)*D(i))];
end
length(unique(x))
for i=1:100000
    t=find(x==x(i));
    if length(t)>1
        t
    end
end
```

References

- [1] James (<https://math.stackexchange.com/users/16811/james>). Why is $f(x) = x\phi(x)$ one-to-one? Mathematics Stack Exchange. URL:<https://math.stackexchange.com/q/68120>, accessed on 15 June 2021.
- [2] P. Prapanpong. Quasi-injectivity of some arithmetic functions. *J. Integer Seq.*, 24, 2021.
- [3] A. Vourdas. *Finite and Profinite Quantum Systems*. Springer, USA, 2017.